

COURSEWORK

7CCSMSDV Introduction to Data Visualization

ABSTRACT

- In this project, Olympic data from 1896 to 2022 is analysed to understand how countries perform over time, whether hosting the Olympics provides an advantage, and which countries dominate specific sports.

1 RESEARCH QUESTIONS

- **Q1: Analyse the development of teams' performances over time. Are there detectable trends?**

This question looks at how countries' performances change over time and whether there are clear patterns such as improvement or decline.

- **Q2: Does being a host country improve a country's medal performance?**

This question checks whether countries perform better when they host the Olympics compared to when they do not.

- **Q3: Which countries dominate each sport and how has this changed over time?**

This question looks at which countries perform best in different sports and whether this dominance changes.

2 DATASET DESCRIPTION

The analysis uses the following datasets:

- **olympic_medals.csv**
This dataset contains information about medals won by countries in different sports and years. It is used to calculate medal counts.
- **olympic_hosts.csv**
This dataset contains information about host countries for each Olympic year. It is used to identify host years.
- **olympic_results.csv**
This dataset contains detailed results of events. It supports sport-level analysis.
- **Paralympic datasets (summer and winter)**

These datasets provide similar structured data and help validate patterns.

These datasets are suitable because they include:

- Year (time)
- Country (who)
- Sport (category)
- Medal count (performance)

3 VISUALIZATION AND ANALYSIS

France, Great Britain, People's Republic of China and United States of America

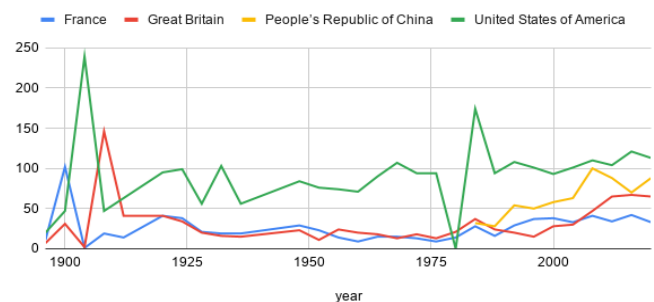


Figure 1.1: Line Chart for Medal Trends Over Time

Stacked Area Chart for Medals Trend Over Time

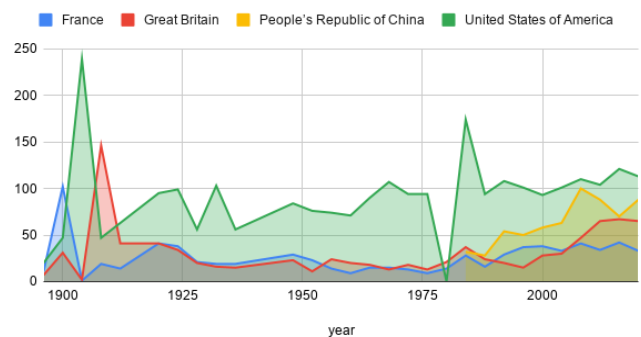


Figure 1.2: Stacked Area Chart for Medal Trends Over Time

Stacked Column Chart for Medal Trends Over Time

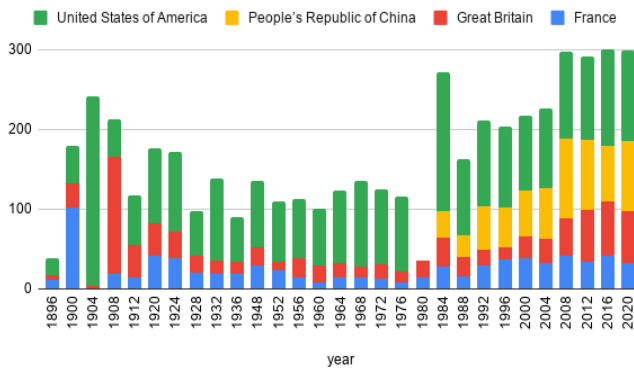


Figure 1.3: Stacked Column Chart for Medal Trends Over Time

Analysis:

These charts illustrate how medal performance has evolved over time for the selected countries. The United States consistently maintains a high medal count across most Olympic years, indicating long-term dominance. China shows a significant rise in performance after the 1980s, reflecting rapid development and increased investment in sports. Great Britain and France display comparatively stable but lower medal counts over time.

The United States consistently dominates across most years, occupying the largest share of the area. However, in later decades, countries like China and Great Britain begin to contribute more significantly, showing a shift toward a more competitive and globally balanced Olympic landscape.

Early Olympics show smaller and irregular totals, while modern Olympics display higher and more consistent medal counts due to increased participation and event categories.

Overall, the chart highlights long-term trends in Olympic performance and reveals how the balance of dominance among countries has shifted across different periods.

Line Chart for Host vs Non Host Analysis

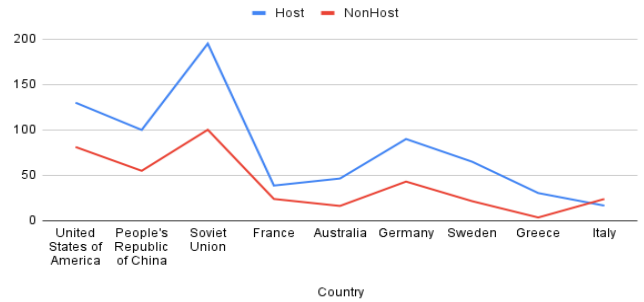


Figure 2.2: Line Chart for Host vs Non-Host Performance

Stacked Column Chart for Host vs Non Host Analysis

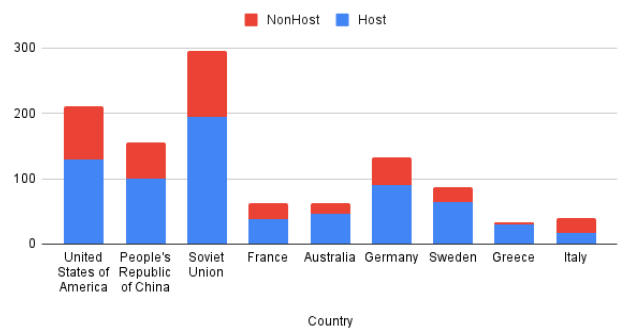


Figure 2.3: Stacked Column Chart for Host vs Non-Host Performance

Analysis:

These charts compare medal performance during host and non-host years across different countries. It shows that most countries achieve higher medal counts when they host the Olympics.

The difference between host and non-host performance is clearly visible for several countries, indicating a consistent host advantage. This improvement may be due to factors such as familiar playing conditions, strong crowd support, and increased investment and preparation leading to the event.

Overall, the charts highlight that hosting the Olympics positively influences a country's performance, although the extent of this advantage varies across countries.

Host and NonHost

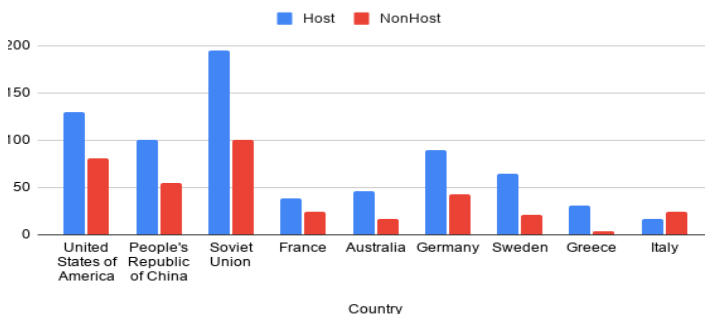


Figure 2.1: Grouped Column Chart for Host vs Non-Host Performance

Sports Dominance of Countries of Time

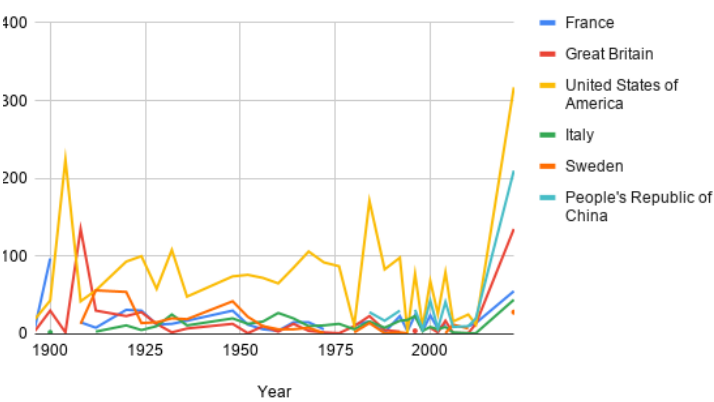


Figure 3.1: Line Chart for Sport Dominance Across Countries

Scatter Plot for Sports Dominance of Countries of Time

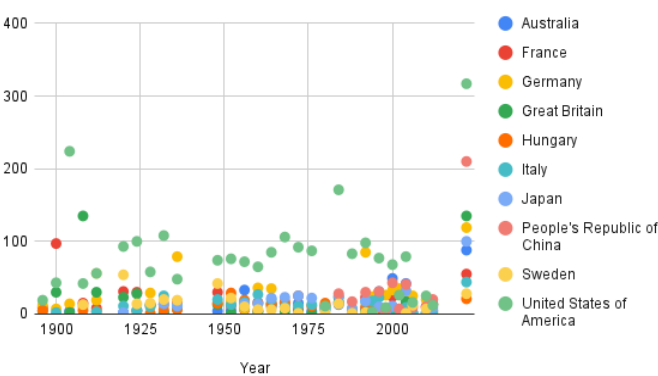


Figure 3.2: Scatter Plot for Sport Dominance Across Countries

Stacked Column Chart for Sports Dominance of Countries of Time

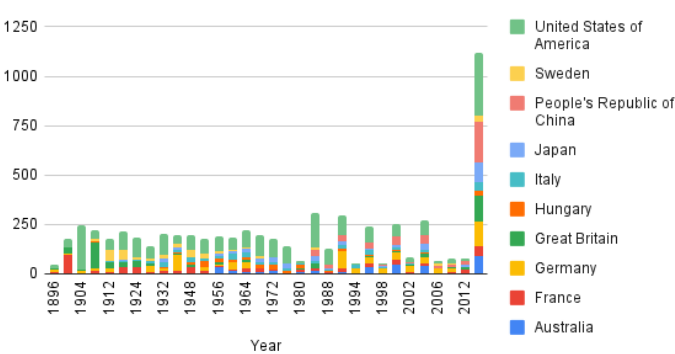


Figure 3.3: Stacked Column Chart for Sport Dominance Across Countries

Analysis:

This chart shows the dominance of the top countries in different sports over time. It highlights how the performance of leading countries changes across Olympic years rather than remaining constant.

Some countries consistently perform strongly across multiple sports. For example, the United States shows sustained dominance in sports like athletics and swimming over several decades. Other countries display periods of rise and decline, indicating shifts in performance and competitiveness.

The United States stands out with consistently higher values and occasional spikes, reflecting strong dominance in multiple years. Other countries such as China and Great Britain show increasing presence in later years, indicating growing competitiveness. Overall, the charts highlight variability in performance and the emergence of new strong contenders over time.

The charts also reflect how certain countries specialize in specific sports, while others maintain overall strength. Overall, it demonstrates that sports dominance evolves over time due to factors such as training, investment, and changing global competition.